

# THE VERIFIER BOTTLENECK

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Can filtering self-generated work  
create new held-out success?

# The question

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Can a verifier turn a model's own samples into useful training data?

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A fixed held-out budget separates learning from merely making known answers easier to sample.

# The task

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Canonical affine-map compositions mod 101.

$$g_i(x) = a_i x + b_i \pmod{101}$$

example:  $g_3(g_1(g_4(x))) \rightarrow \text{ANSWER} = (A, B)$

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10,100 possible full-map answers. Exact check.

# The experiment

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Hold everything fixed. Change two things.



Every comparison starts from the same saved candidate pool.

# The controls

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Same model. Same saved samples. Same training budget.

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BASE

no update

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ORACLE

gold SFT

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EXACT

exact filter

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RANDOM

random filter

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All labels use held-out pass@16 at a fixed 16-sample test budget.

# The outcome

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At fixed  $K_{\text{eval}} = 16$ :

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## EXPANSION

pass@16 rises;  
net gain > loss

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## SHARPENING

pass@1 rises;  
pass@16 is flat

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## COLLAPSE

pass@16  
falls

Predeclared paired uncertainty; anything else is inconclusive.

# Four next steps

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## 1. Agree

Task + split

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Model + compute

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Controls + metric

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Success + stop rule

Then

2. PREPARE

3. SKETCH

4. REVIEW



docs +  
manifests

one clear  
script

everyone  
reviews v1

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Freeze together, then calibrate and run the core comparison.